

3 Simultaneous generation of judgements

When stating proof rules, judgements may be mixed and matched to generate more complicated evidence. For example, the following rules generate judgements on the parity of natural numbers.

$$\begin{array}{ccc} \frac{\text{EVENZ}}{\text{zero even}} & \frac{\text{ODD} \quad n \text{ even}}{\text{succ}(n) \text{ odd}} & \frac{\text{EVEN} \quad n \text{ odd}}{\text{succ}(n) \text{ even}} \end{array}$$

4 Parallels with functional programming

Notice that the above sets of rules have a flavour akin to the following data type definitions in Haskell.

```
data Nat = Zero | Succ Nat

data Even = Zero | Succ Odd
data Odd = Succ Even
```

This correspondence is rather deep, and leads to modern **proof assistants**.